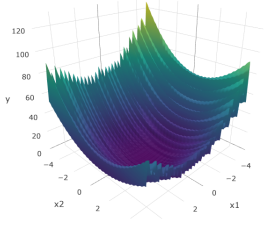
Optimization in Machine Learning

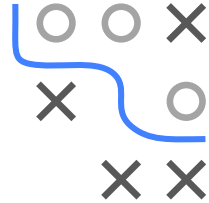
Multi-Start Optimization



Learning goals

- Multimodal functions
- Basins of Attractions
- Simple multi-start procedure

MOTIVATION



- So far: derivative-free methods for *unimodal* objective function (exception: simulated annealing)
- With multimodal objective functions, methods converge to **local minima**.
- Optimum found may differ for different starting values $\mathbf{x}^{[0]}$

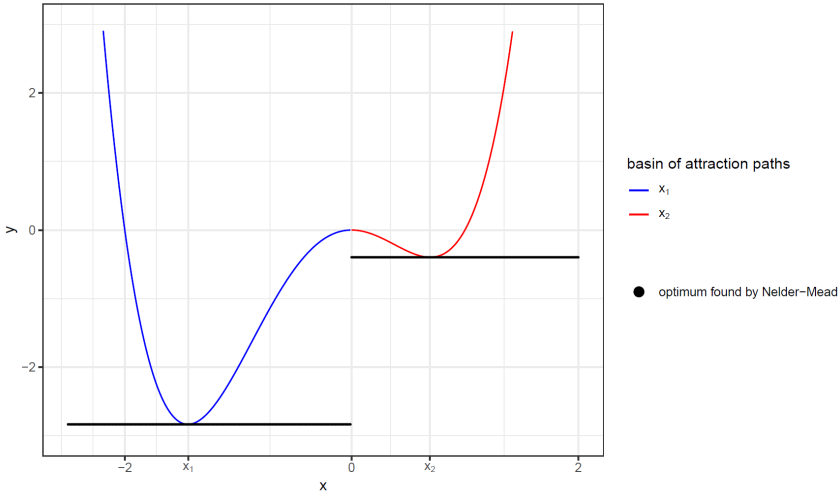
Attraction areas:

- Let $f_1^*, \ldots, f_k^*$ be local minimum values of f with $f_i^* \neq f_j^* \quad \forall i \neq j$.
- Notation: $A(\mathbf{x}^{[0]})$ denotes result of algorithm A started at $\mathbf{x}^{[0]}$
- Then: Set

$$\mathcal{A}(f_i^*, A) = \{\mathbf{x} : A(\mathbf{x}) = f_i^*\}$$

  is called *attraction area*/*basin of attraction* of f_i^* for algorithm A

ATTRACTION AREAS

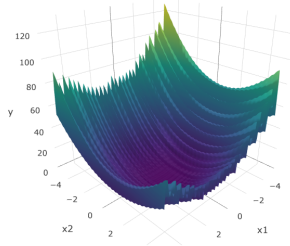


MULTI-STARTS

Levy function:

$$f(\mathbf{x}) = \sin^2(3\pi x_1) + (x_1 - 1)^2[1 + \sin^2(3\pi x_2)] + (x_2 - 1)^2[1 + \sin^2(2\pi x_2)]$$

- Global minimum: $f(\mathbf{x}^*) = 0$ at $\mathbf{x}^* = (1, 1)^\top$
- Optimize f by BFGS method with random starting point in $[-2, 2]^2$ and collect result
- Repeat 100 times



Distribution of results (y values):

```
## Min.     1st Qu.    Median    Mean     3rd Qu.    Max.
## 0.0000   0.1099     0.5356    2.4351   1.9809     18.3663
```